

How the Ensemble Model Calculates Its Signal

The Ensemble is not a single model - it's a two-layer decision system (meta-learner). Here is exactly how it works, step by step.

Layer 0 - The 5 Worker Models

These all run independently and each gives a vote + confidence score:

- Wavelet: Filters noise from price data using wave math, detects trend (Best in: Any market)
- HMM: Detects which 'hidden state' (regime) the market is in (Best in: Transitions)
- LSTM: Deep learning; predicts next move from sequences of past prices (Best in: Trending)
- TFT: Forecasts multiple steps ahead simultaneously (Best in: Growth)
- Genetic: Evolves trading rules via natural selection (Best in: Any)

Each produces something like: {signal: 'LONG', confidence: 0.82}

Layer 1 - The Ensemble Combines Them

Mode A: ML Meta-Learner (when trained on 50+ historical trades)

Feature vector includes:

- direction x confidence for each model
- raw confidence for each model
- regime code (GROWTH=+1, NORMAL=0, CRISIS=-1)
- model agreement score

Mode B: Dynamic Regime Weighting (the current live mode)

When there's not enough training data, it uses a weighted vote:

1. Assign regime-based weights (e.g. GROWTH: wavelet:25%, ensemble:50%...)
2. Performance adaptation (after 10+ trades) - Models that WON recently get heavier votes.
3. Agreement bonus/penalty - If 4 or more models agree, they get a 15% boost.
4. Weighted vote - score = weight x confidence
5. Disagreement penalty on final confidence
6. Output - If best_score < 0.25 -> HOLD, else -> best direction.

Layer 2 - Dynamic Weight Adjustments

After every closed trade, the DynamicWeightAdjuster updates each model's weight for next time:

$$\text{new_weight} = \text{normalize}(\text{regime_base} \times \text{performance_multiplier} \times \text{agreement_factor})$$

The performance multiplier is based on rolling Sharpe ratio (last 50 trades):

- A model that's been winning lately -> Sharpe goes up -> gets more weight

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- A model that's been losing -> Sharpe goes down -> weight shrinks

This is exactly how Renaissance Technologies operates - the models that have been right recently are trusted more, and the weights constantly adapt.

In Plain English

Every 60 seconds, Wavelet + HMM + LSTM + TFT + Genetic all look at the gold price from their own angle and vote LONG/SHORT/HOLD. The Ensemble collects all the votes, weights them by who has been winning lately and what the current market regime is, applies a penalty if models disagree, and produces one final signal with a confidence score. If confidence > 60%, the paper trading engine executes the trade.